Consecutive-Cycle Schreier Graphs

Slides from Section 4 onward of `f2-p6-cck.tex`

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f2-p5-cck-slides-fromp24.tex

Starting point: from tree token moves to cycles

Bridge from the tree formula

The path P_n has sites $1, \ldots, n$ and cuts between adjacent positions. The suffix heights η_i are the tree heights of P_n when zeros are the tokens.

- $p = 2$: generators are adjacent transpositions; the tree formula is exact.
- $p \geq 3$: a consecutive p-cycle is a non-tree shortcut spanning $p - 1$ cuts.
- The cut-area remains a lower bound, but a residual correction appears.

$$A := 0^m 1^k, \qquad m + k = n.$$

State space and action convention

Definition (Binary state space)

$$V_{m,n} := \{X \in \{0,1\}^n : \#\{j : x_j = 0\} = m\}, \qquad Z(X) := \{j : x_j = 0\}.$$

A permutation $\sigma \in S_n$ acts on positions by sending the symbol at j to position $\sigma(j)$:

$$(\sigma X)_t = x_{\sigma^{-1}(t)}.$$

For a consecutive p-cycle,

$$\gamma_i^{(p)} = (i\ i+1\ \cdots\ i+p-1), \qquad 1 \leq i \leq n-p+1.$$

Components as group orbits

Let

$$H_p(n) := \langle \gamma_i^{(p)} : 1 \leq i \leq n-p+1 \rangle \leq S_n.$$

Proposition (Components as orbits)

The connected components of $G_p(m,n)$ are precisely the orbits of $H_p(n)$ on m-element subsets of $[n]$, equivalently on $V_{m,n}$.

Canonical component

For residual statistics we write

$$\mathcal{C}_p(A)$$

for the connected component of $A = 0^m 1^k$.

Suffix heights and cut-area

For the cut between positions i and $i+1$,

$$\eta_i(X) := |Z(X) \cap \{i+1, \ldots, n\}| = \sum_{j=i+1}^{n} (1 - x_j), \qquad 1 \leq i \leq n-1.$$

Definition (Cut-area)

$$\mathrm{Area}_{\mathrm{cut}}(X, Y) := \sum_{i=1}^{n-1} |\eta_i(X) - \eta_i(Y)|.$$

This records the total number of required cut crossings, ignoring whether they can be packed into legal p-cycle moves.

Cut-area as a footrule distance

Let $X, Y \in V_{m,n}$ have ordered zero positions

$$\alpha_1 < \cdots < \alpha_m, \qquad \beta_1 < \cdots < \beta_m.$$

Lemma (Cut-area footrule identity)

$$\mathrm{Area}_{\mathrm{cut}}(X, Y) = \sum_{r=1}^{m} |\alpha_r - \beta_r|.$$

Meaning

Rank-matched zero displacement equals the L^1 distance between the corresponding suffix counting functions.

One window changes at most $p-1$ cuts

A generator supported on

$$\{a, a+1, \ldots, a+p-1\}$$

changes only the internal cuts

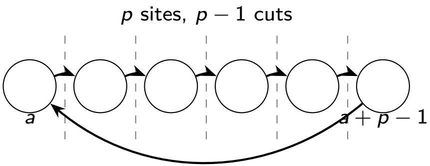
$$a, a+1, \ldots, a+p-2.$$

For each affected cut,

$$|\eta_i(\sigma X) - \eta_i(X)| \leq 1.$$

Therefore

$$\sum_i |\eta_i(\sigma X) - \eta_i(X)| \leq p-1.$$



$G_2(m, n)$: the exact path case

The consecutive 2-cycle generators are adjacent transpositions

$$s_i = (i\ i+1), \qquad 1 \le i \le n-1.$$

Proposition (Exact formula)

For all $X, Y \in V_{m,n}$,

$$d_{G_2}(X, Y) = \mathrm{Area}_{\mathrm{cut}}(X, Y).$$

Proposition (Diameter)

$$\mathrm{diam}(G_2(m, n)) = mk.$$

So the G_2 residual is identically zero:

$$h_2(X, Y) = 0.$$

$G_3(m,n)$: definitions

The consecutive 3-cycle generators are

$$\gamma_i = (i\ i+1\ i+2), \qquad 1 \le i \le n-2.$$

They generate A_n, and $G_3(m,n)$ is connected for all admissible m,n.
For B with zero positions

$$1 \le z_1 < z_2 < \cdots < z_m \le n,$$

define

$$d_i := z_i - i, \qquad I(B) := \sum_{i=1}^{m} d_i,$$

$$\Delta(B) := \left| \sum_{i=1}^{m} (-1)^{i+1} (d_i \bmod 2) \right|.$$

G_3: canonical distance formula

Theorem (Canonical G_3 formula)

For all $B \in V_{m,n}$,

$$d_{G_3(m,n)}(A,B) = \frac{I(B)+\Delta(B)}{2} = \left\lceil \frac{I(B)}{2} \right\rceil + \left\lfloor \frac{\Delta(B)}{2} \right\rfloor .$$

Equivalently,

$$h_3(B) := d_{G_3}(A,B) - \left\lceil \frac{I(B)}{2} \right\rceil = \left\lfloor \frac{\Delta(B)}{2} \right\rfloor .$$

Interpretation

The area term is holographic; the only remaining canonical G_3 correction is a parity imbalance.

G_3 parity correction: a small example

Let $m = 3$, $n = 6$, and $A = 000111$. Take

$$B = 101010, \qquad Z(B) = (2, 4, 6).$$

Then

$$(d_1, d_2, d_3) = (1, 2, 3), \qquad I(B) = 6, \qquad \Delta(B) = |1 - 0 + 1| = 2.$$

The cut-area lower bound gives only

$$d_{G_3}(A, B) \geq \left\lceil \frac{6}{2} \right\rceil = 3,$$

but the exact formula gives

$$d_{G_3}(A, B) = \frac{6 + 2}{2} = 4.$$

A shortest path is

$$000111 \to 010011 \to 100011 \to 101001 \to 101010.$$

G_3: arbitrary-pair two-sided bounds

Let X, Y have ordered zero positions α_i and β_i.

Theorem (Two-sided bounds)

$$\left\lceil \frac{1}{2}\sum_{i=1}^{m}|\alpha_i - \beta_i| \right\rceil \leq d_{G_3(m,n)}(X, Y) \leq \sum_{i=1}^{m}\left\lceil \frac{|\alpha_i - \beta_i|}{2} \right\rceil.$$

Move catalog summary

Each nontrivial G_3 move is one of:

- one zero moves by ± 2;
- two adjacent zeros both move by $+1$ or both by -1;
- one zero moves by ± 1.

Thus one move changes the sorted zero-position vector by total L^1 at most 2.

G_3: eccentricity and diameter conjecture

Proposition (Canonical eccentricity)

For $k = n - m$,

$$\mathrm{ecc}_{G_3(m,n)}(A) = \left\lceil \frac{mk}{2} \right\rceil .$$

The maximum is attained at $A^* = 1^k 0^m$ and is unique iff m and k are both odd.

Conjecture (Diameter)

$$\mathrm{diam}(G_3(m,n)) = \left\lceil \frac{m(n-m)}{2} \right\rceil .$$

Verified by exhaustive BFS over all pairs for $3 \leq n \leq 9$.

G_3 residual levels

Definition

$$h_3(X,Y) := d_{G_3}(X,Y) - \left\lceil \frac{\mathrm{Area}_{\mathrm{cut}}(X,Y)}{2} \right\rceil .$$

Conjecture (Residual-level counts)

For fixed m and $j \geq 0$,

$$H_{3,j}(m,n) := \#\{B \in \mathcal{C}_3(A) : h_3(B) = j\}$$

is eventually a quasi-polynomial in n with period dividing 2.

Observed first cases:

$$H_{3,1}(3,n) = \binom{\lfloor n/2 \rfloor}{3}, \quad H_{3,2}(7,n) = \binom{\lfloor n/2 \rfloor}{7}, \quad H_{3,3}(11,n) = \binom{\lfloor n/2 \rfloor}{11}.$$

The consecutive 4-cycle generators are

$$\gamma_i = (i\ i+1\ i+2\ i+3), \qquad 1 \leq i \leq n-3.$$

Proposition (G_4 cut-area lower bound)

For X, Y in the same connected component,

$$d_{G_4}(X, Y) \geq \left\lceil \frac{\mathrm{Area}_{\mathrm{cut}}(X, Y)}{3} \right\rceil.$$

Definition (G_4 residual)

$$h_4(X, Y) := d_{G_4}(X, Y) - \left\lceil \frac{\mathrm{Area}_{\mathrm{cut}}(X, Y)}{3} \right\rceil \geq 0.$$

G_4: local five-site window diameter

Consider the local graph $G_4(q,5)$ generated by

$$a = (1\ 2\ 3\ 4), \qquad b = (2\ 3\ 4\ 5),$$

and inverses.

Lemma (Local diameter)

For $1 \le q \le 4$*,*

$$\operatorname{diam}(G_4(q,5)) \le 3.$$

Equality occurs for $q = 2$ *and, by complementation, for* $q = 3$*.*

Use later

This gives the local constant $D_4^{\mathrm{loc}} = 3$ used to simulate adjacent swaps in larger $G_4(m,n)$ graphs.

G_4: exact formula for one zero

Let $m = 1$, and let B have its unique zero at position z. Write

$$z - 1 = 3q + r, \qquad r \in \{0, 1, 2\}.$$

Proposition (One-particle formula)

$$d_{G_4}(A, B) = q + r.$$

Equivalently, $h_4(B) = 1$ exactly when $z \equiv 0 \pmod 3$, and $h_4(B) = 0$ otherwise.

Smallest non-sharp lower-bound example: if $z = 3$, then

$$\left\lceil \frac{I(B)}{3} \right\rceil = 1, \qquad d_{G_4}(A, B) = 2.$$

G_4: small-m conjectural formulas

Let $A = 0^m 1^k$, let B have zero displacements $a_i = z_i - i$, and residues $r_i = a_i \bmod 3$.

Conjecture (Small-m formulas)

For $m = 2$,

$$d_{G_4}(A, B) = \left\lceil \frac{a_1 + a_2 + (a_1 \bmod 3)}{3} \right\rceil .$$

For $m = 3$ and $k \geq 4$,

$$d_{G_4}(A, B) = \left\lceil \frac{I(B)}{3} \right\rceil + \mathbf{1}_{\{r_1, r_2, r_3\} = \{0,1,2\}} .$$

These are computationally supported patterns, not proved statements.

G_4: a small $m = 2$ residual example

Let $m = 2$, $n = 5$, and $A = 00111$. Take

$$B = 10101, \qquad Z(B) = (2, 4).$$

Then

$$(a_1, a_2) = (1, 2), \qquad I(B) = 3.$$

The lower bound gives

$$d_{G_4}(A, B) \geq \left\lceil \frac{3}{3} \right\rceil = 1,$$

but the conjectural $m = 2$ formula gives

$$d_{G_4}(A, B) = \left\lceil \frac{1 + 2 + 1}{3} \right\rceil = 2.$$

Realizing path:

$$00111 \to 01011 \to 10101.$$

General $G_p(m,n)$: lower and upper bounds

For $p \geq 2$, use

$$\mathcal{S}_p(n) = \{\gamma_i^{(p)}, (\gamma_i^{(p)})^{-1} : 1 \leq i \leq n-p+1\}.$$

Proposition (General cut-area lower bound)

$$d_{G_p}(X,Y) \geq \left\lceil \frac{\mathrm{Area}_{\mathrm{cut}}(X,Y)}{p-1} \right\rceil.$$

Proposition (Coarse local upper bound, $n \geq p+1$)

If D_p^{loc} is the maximum diameter of the local $(p+1)$-site graph $G_p(q, p+1)$, then

$$d_{G_p}(X,Y) \leq D_p^{\mathrm{loc}} \sum_{r=1}^{m} |\alpha_r - \beta_r|.$$

Also

$$D_p^{\mathrm{loc}} \leq \binom{p+1}{\lfloor (p+1)/2 \rfloor} - 1.$$

Local constants D_p^{loc}

Exhaustive BFS over local layers gives:

p	2	3	4	5	6	7	8
D_p^{loc}	2	2	3	4	7	8	10
$\binom{p+1}{\lfloor (p+1)/2 \rfloor} - 1$	2	5	9	19	34	69	125

- The binomial bound is attained only at $p = 2$ in this range.
- The tempting pattern $D_p^{\mathrm{loc}} = p - 1$ holds for $3 \leq p \leq 5$ but fails at $p = 6$.
- Witness: in $G_6(3, 7)$, 1100011 and 1010101 are at distance 7.

G_4: sharper arbitrary-pair bounds

Assume $n \geq 5$ and let $X, Y \in V_{m,n}$.

Corollary (Two-sided G_4 bounds)

$$\max\left\{\left\lceil \frac{\mathrm{Area}_{\mathrm{cut}}(X,Y)}{3} \right\rceil, \left\lceil \frac{d_{\mathrm{Ham}}(X,Y)}{4} \right\rceil\right\} \leq d_{G_4(m,n)}(X,Y) \leq 3\sum_{r=1}^{m} |\alpha_r - \beta_r|.$$

Since $\sum_r |\alpha_r - \beta_r| = \mathrm{Area}_{\mathrm{cut}}(X,Y)$,

$$d_{G_4(m,n)}(X,Y) \leq 3\,\mathrm{Area}_{\mathrm{cut}}(X,Y).$$

In particular, $G_4(m,n)$ is connected for all $n \geq 5$.

Canonical area statistic

For $A = 0^m 1^k$ and B with zero positions

$$1 \leq z_1 < \cdots < z_m \leq n,$$

define

$$I(B) := \sum_{a=1}^{m} (z_a - a).$$

Proposition

$$\mathrm{Area}_{\mathrm{cut}}(A, B) = I(B).$$

Combining with the general lower bound,

$$d_{G_p}(A, B) \geq \left\lceil \frac{I(B)}{p-1} \right\rceil.$$

The canonical residual is

$$h_p(B) := d_{G_p}(A, B) - \left\lceil \frac{I(B)}{p-1} \right\rceil.$$

Residual conjectures

Conjecture (Quasi-polynomial residual counts)

Fix $p \geq 3$, $m \geq 1$, and $j \geq 0$. For all sufficiently large n,

$$H_{p,j}(m,n) := \#\{B \in \mathcal{C}_p(A) : h_p(B) = j\}$$

agrees with a quasi-polynomial in n whose period divides $p-1$.

Conjecture (Finite local correction)

For fixed p and m, the residual $h_p(B)$ is governed by finitely many local obstruction types after subtracting the leading term

$$\left\lceil \frac{I(B)}{p-1} \right\rceil .$$

Proof idea for the G_3 formula: lower bound

For a state X, define

$$I(X) = \sum_i d_i(X), \qquad S(X) = \sum_i (-1)^{i+1}(d_i(X) \bmod 2), \qquad \Delta(X) = |S(X)|.$$

For $\varepsilon \in \{+1, -1\}$, use the potential

$$\Phi_\varepsilon(X) := I(X) + \varepsilon S(X).$$

One G_3 move increases Φ_ε by at most 2. Choosing $\varepsilon S(B) = \Delta(B)$ gives, for any path of length L,

$$I(B) + \Delta(B) = \Phi_\varepsilon(B) - \Phi_\varepsilon(A) \leq 2L.$$

Hence

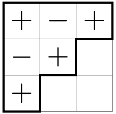
$$d_{G_3}(A, B) \geq \frac{I(B) + \Delta(B)}{2}.$$

Proof idea for the G_3 formula: Ferrers removal

Draw the displacement vector $0 \le d_1 \le \cdots \le d_m$ as a Ferrers diagram. Color the cell in row i, column j by $(-1)^{i+j}$.

Allowed removals toward A:

- horizontal domino;
- vertical domino;
- one majority-color boundary cell.



$d = (1, 2, 3)$ gives $S = 2$, $\Delta = 2$.

Each legal removal decreases

$$F(X) := \frac{I(X) + \Delta(X)}{2}$$

by exactly 1.

Induction on $I(B)$ gives the matching upper bound.

Exact statements versus conjectural directions

Family	Proved statement	Residual question
Tree token graph	$d(X,Y) = \sum_{e \in E(T)} \lvert \eta_e(X) - \eta_e(Y) \rvert$	No residual.
$G_2(m,n)$	$d_{G_2}(X,Y) = \mathrm{Area}_{\mathrm{cut}}(X,Y)$, $\mathrm{diam}(G_2) = mk$	$h_2 \equiv 0$.
$G_3(m,n)$	$d(A,B) = (I + \Delta)/2$ and arbitrary-pair bounds	Classify residual levels and prove diameter conjecture.
$G_4(m,n)$	Cut-area/Hamming lower bound; $3\,\mathrm{Area}_{\mathrm{cut}}$ upper bound for $n \geq 5$	Sharpen bounds; characterize h_4.
General G_p	$d \geq \lceil \mathrm{Area}_{\mathrm{cut}}/(p-1) \rceil$ and local-diameter upper bound	Quasi-polynomial residual levels.

What the result does not yet cover

- A full connectivity classification for all $G_p(m, n)$.
- A general closed formula for h_p when $p \geq 4$.
- A proof of the arbitrary-pair diameter conjecture for $G_3(m, n)$.
- Sharp universal upper bounds for G_4 beyond the local-diameter estimate.
- A theorem explaining why residual-level counts should be quasi-polynomial.

Main open target

Replace exact search for $h_4, h_5, \ldots$ by finite local obstruction formulas.

G_4 distance computation: cut-area heuristic

For a fixed target Y, define

$$\widehat{d}_4(S, Y) := \left\lceil \frac{\text{Area}_{\text{cut}}(S, Y)}{3} \right\rceil.$$

By the G_4 lower bound,

$$\widehat{d}_4(S, Y) \leq d_{G_4}(S, Y).$$

The same one-window estimate also implies consistency:

$$\widehat{d}_4(S, Y) \leq 1 + \widehat{d}_4(S', Y)$$

whenever S and S' are adjacent.

Consequence

A^*, bidirectional A^*, and IDA* with $\widehat{d}_4$ remain exact.

A^* and IDA* for G_4

For a search from X to Y,

$$f(S) = g(S) + \widehat{d}_4(S, Y),$$

where $g(S)$ is distance already traveled from X to S.

Algorithm	Advantage	Disadvantage
BFS	Simple and exact	Huge memory use
A^*	Exact; often far fewer states	Can still use large memory
IDA*	Exact; low memory	May repeat work

Initial IDA* cutoff:

$$C_0 = \widehat{d}_4(X, Y) = \left\lceil \frac{\mathrm{Area}_{\mathrm{cut}}(X, Y)}{3} \right\rceil .$$

Reproducibility protocol

For fixed (p, m, n):

```
k = n - m
A = 0^m 1^k
generators = {gamma_i^(p), inverse(gamma_i^(p))}
queue = [A]
dist[A] = 0
while queue is nonempty:
    S = pop_left(queue)
    for sigma in generators:
        T = sigma(S)
        if T != S and T not in dist:
            dist[T] = dist[S] + 1
            push_right(queue,T)
for each reached state B:
    compute zero positions z_1 < ... < z_m
    I(B) = sum_a (z_a - a)
```

Finite computational evidence

- The G_3 exact formula was checked for $m = 1, 2, 3, 4$ over small n ranges.
- The conjectural G_4 formulas were checked by exhaustive BFS:

$$m = 2,\ n = 4\text{–}11, \qquad m = 3,\ n = 7\text{–}12.$$

- Higher residual-level checks agree with predicted binomial formulas:

$$H_{3,3}(11, n): \quad n = 22, 23, 24, 26,$$

$$H_{4,2}(5, 19) = 6, \qquad H_{4,3}(8, 26) = 1, \qquad H_{4,3}(8, 27) = 9.$$

Status

These computations support the conjectures but are not used in the proofs.

Verified residual-level patterns

G_3 first visible levels

$$H_{3,1}(3, n) = \binom{\lfloor n/2 \rfloor}{3}$$

$$H_{3,2}(7, n) = \binom{\lfloor n/2 \rfloor}{7}$$

$$H_{3,3}(11, n) = \binom{\lfloor n/2 \rfloor}{11}$$

G_4 first higher levels

$$H_{4,2}(5, n) = \binom{\lfloor n/3 \rfloor}{5}$$

$$H_{4,3}(8, n) = \binom{\lfloor n/3 \rfloor}{8}$$

Each higher-residual binomial prediction is supported by at least two nonzero data points.

Proof verification summary

The proof of

$$d_{G_3(m,n)}(A, B) = \frac{I(B) + \Delta(B)}{2}$$

has three verified components:

1. **Lower bound:** potential $\Phi_\varepsilon = I + \varepsilon S$ changes by at most 2 per move.
2. **Removable boundary piece:** every nonzero Ferrers diagram has a legal domino or majority-cell removal.
3. **Constructive upper bound:** induction on $I(B)$ using removals that decrease $(I + \Delta)/2$ by 1.

The parity identity

$$I \equiv S \equiv \Delta \pmod 2$$

then gives the equivalent residual form

$$h_3(B) = \left\lfloor \frac{\Delta(B)}{2} \right\rfloor .$$

Notation cheat sheet

Symbol	Meaning
$V_{m,n}$	binary strings of length n with m zeros
$Z(X)$	zero set of state X
$A = 0^m 1^k$	canonical state, $m + k = n$
$\gamma_i^{(p)}$	consecutive p-cycle on sites $i, \dots, i + p - 1$
$G_p(m, n)$	Schreier graph generated by consecutive p-cycles
$\eta_i(X)$	suffix height across cut $i \vert i + 1$
$\mathrm{Area}_{\mathrm{cut}}(X, Y)$	$\sum_i \lvert\eta_i(X) - \eta_i(Y)\rvert$
$I(B)$	canonical area $\sum_a (z_a - a)$
$\Delta(B)$	G_3 parity imbalance
$h_p(X, Y)$	holographic residual above the cut-area lower term

Takeaway

Core message

For consecutive-cycle Schreier graphs, the tree/path cut-area remains the leading "holographic" term, but non-tree generators create residual corrections.

- G_2: cut-area is exact.
- G_3: residual is exactly parity: $h_3 = \lfloor \Delta/2 \rfloor$ from A.
- G_4 and beyond: residuals are empirical/local-obstruction phenomena awaiting closed formulas.